

Balancing act

A

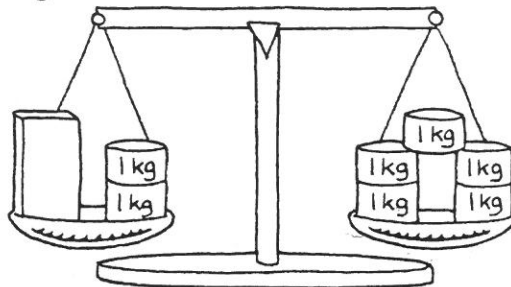
These scales show a parcel and some kilogram weights.

To find the mass (m) of the parcel, you can use the equation: $m + 2 = 5$

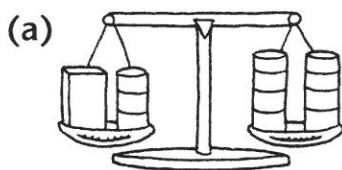
To find the answer, subtract 2 weights from both sides. The scales will still balance!

$$\begin{aligned} m + 2 - 2 &= 5 - 2 \\ m &= 3 \end{aligned}$$

So the mass of the parcel is 3 kilograms.

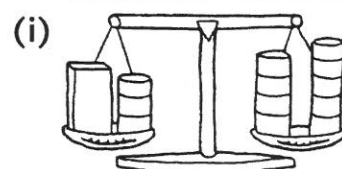
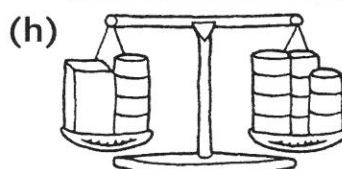
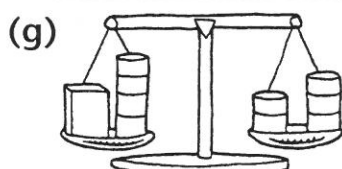
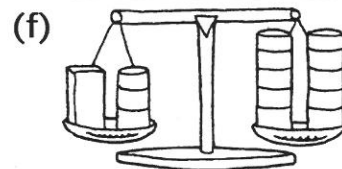
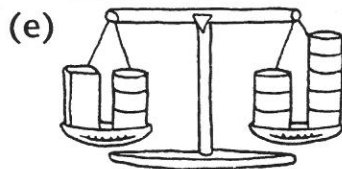
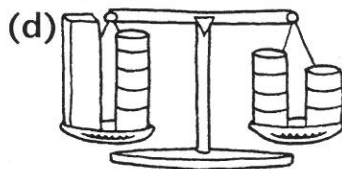
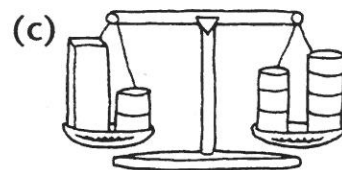
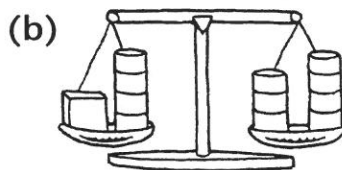


1. Write an equation for each set of scales. Solve it to find the mass of the parcel.



$$m + 3 = 8$$

$$m = 5$$



2. Find the value of the letter in each of these equations.

(a) $y + 7 = 24$

(b) $c + 14 = 32$

(c) $16 + t = 32$

$y =$ _____

B

You can add the same amount to both sides and the scales will still balance. Solve these equations by adding the same to both sides.

(a) $n - 5 = 20$

(b) $n - 9 = 14$

(c) $n - 7 = 14$

(d) $n - 13 = 22$

$n =$ _____

(e) $n - 56 = 100$

(f) $n - 42 = 13$

(g) $n - 45 = 18$

(h) $n - 17 = 26$



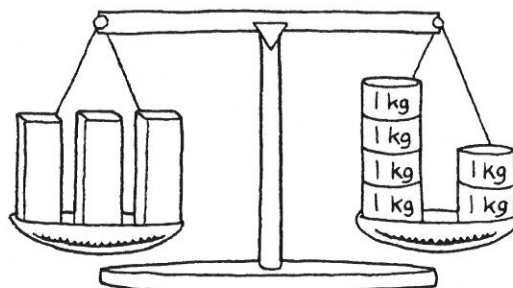
The equals sign tells you that what is on one side of the sign is equal to (or balances) what is on the other side, for example $5 + 9 = 6 + 8$ (both sides have the same value, 14). Check each of your answers by putting the missing number into the question.

Balancing act

Solve simple linear equations using ideas of balance

C

Think of an equation as being on a set of scales. You can add or subtract any number, and multiply or divide by any number. As long as you do exactly the same to both sides of the equation, the scales will still balance. These scales show the equation: $3m = 6$
Divide both sides by 3 to solve the equation:
 $m = 2$



1. Solve these equations by doing the same to both sides.

(a) $m + 6 = 13$

$m = 7$

(b) $5m = 35$

(c) $m - 11 = 21$

(d) $4m = 80$

(e) $7m = 56$

(f) $m + 16 = 40$

(g) $m - 30 = 42$

(h) $m \div 5 = 4$

(i) $8m = 48$

(j) $m + 15 = 28$

(k) $9m = 81$

(l) $6m = 24$

Sometimes you need to use two operations.

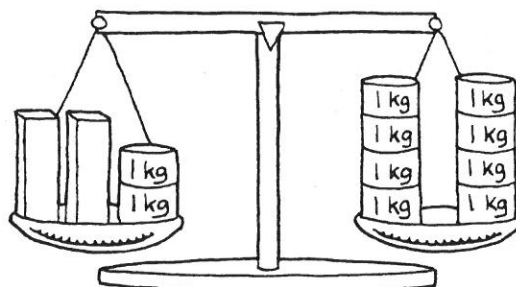
$2m + 2 = 8$

First subtract 2 from both sides:

$2m = 6$

Now divide both sides by 2:

$m = 3$



2. Solve these equations.

(a) $2m + 6 = 16$

$2m = 10$

$m = 5$

(b) $5m + 4 = 19$

(c) $5m + 7 = 37$

(d) $4m + 4 = 12$

(e) $7m + 3 = 38$

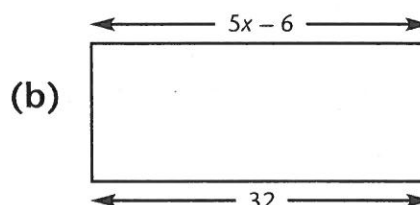
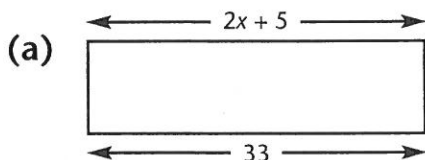
(f) $4m + 16 = 60$

(g) $2m - 10 = 10$

(h) $6m - 6 = 18$

NOW TRY THIS!

• Find the value of x for each rectangle.



• Make up your own rectangle puzzle for a partner to solve.



The equals sign tells you that what is on one side of the sign is equal to (or balances) what is on the other side, for example $5 + 9 = 6 + 8$ (both sides have the same value, 14). Check each of your answers by putting the missing number into the question.

Let-ters begin!

A

In each calculation, the letter on the flap stands for an **unknown number**. Write which number the letter stands for.

$$3 + \boxed{a} = 7 \quad a = \underline{4}$$

$$5 \times \boxed{b} = 15 \quad b = \underline{3}$$

$$9 - \boxed{c} = 2 \quad c = \underline{\quad}$$

$$14 + \boxed{d} = 36 \quad d = \underline{\quad}$$

$$\boxed{e} - 12 = 11 \quad e = \underline{\quad}$$

$$\boxed{f} \div 3 = 6 \quad f = \underline{\quad}$$

$$\boxed{g} \times 9 = 36 \quad g = \underline{\quad}$$

$$25 \div \boxed{h} = 5 \quad h = \underline{\quad}$$

$$49 \div \boxed{i} = 7 \quad i = \underline{\quad}$$

$$\boxed{j} + \boxed{j} = 24 \quad j = \underline{\quad}$$

$$\boxed{k} \div 9 = 9 \quad k = \underline{\quad}$$

$$11 \times \boxed{l} = 66 \quad l = \underline{\quad}$$

$$32 + \boxed{m} = 50 \quad m = \underline{\quad}$$

$$100 - \boxed{n} = 46 \quad n = \underline{\quad}$$

$$\boxed{o} \div 5 = 20 \quad o = \underline{\quad}$$

$$\boxed{p} \times 8 = 32 \quad p = \underline{\quad}$$

$$50 \div \boxed{q} = 10 \quad q = \underline{\quad}$$

B

In these expressions, the letter n stands for **any number**.

1. Join each speech bubble to its matching expression.



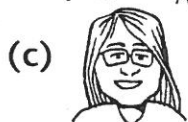
Think of a number and add 4.

$$4 - n$$



Think of a number and subtract it from 4.

$$n + 4$$



Think of a number, divide it by 4 and add 4.

$$n \times 4 - 4$$



Think of a number, multiply it by 4 and subtract 4 from it.

$$n \times 4 + 4$$

$$n + 4 \div 4$$

2. For each speech bubble, choose any number you like to stand for n . Write the answer.

Answer

Answer

Answer

Answer

(a) $n = \underline{3} \quad \underline{7}$ (b) $n = \underline{\quad} \quad \underline{\quad}$ (c) $n = \underline{\quad} \quad \underline{\quad}$ (d) $n = \underline{\quad} \quad \underline{\quad}$

3. On the back of this sheet, write five speech bubbles and matching expressions of your own.

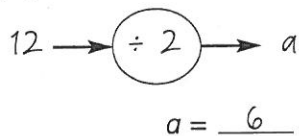


In algebra, letters are used in different ways. Sometimes a letter stands for an **unknown number** that you have to find: for example, $n + 2 = 5$ (n stands for the number 3). Sometimes a letter stands for **any number** you choose: for example, $n + 3 = ?$ (n could be any number, such as 0, 6, 200, -8, 0.5 and so on).

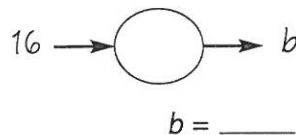
A

1. Use **inverse** operations to solve these equations. Work back through the question from the answer to find the missing number.

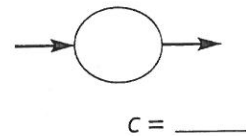
(a) $a \times 2 = 12$



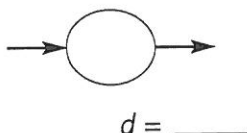
(b) $b + 5 = 16$



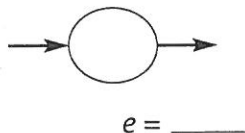
(c) $c - 9 = 25$



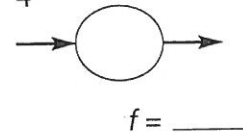
(d) $d \div 6 = 8$



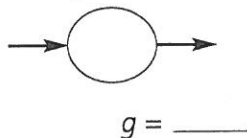
(e) $5 \times e = 35$



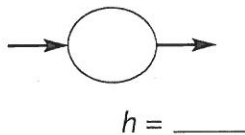
(f) $\frac{f}{4} = 8$



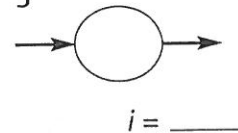
(g) $17 + g = 32$



(h) $6h = 42$



(i) $\frac{i}{3} = 6$



2. Check each of your answers by putting the missing number into the question.

B

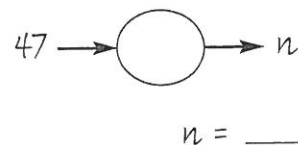
1. Write an equation for each situation. Use inverse operations to find the unknown number.

(a)

I think of a number and add 9.
The answer is 47.
What is my number?



$n + 9 = 47$



(b)

I think of a number and multiply it by 25. The answer is 175.
What is my number?



$25n = 175$

(c)

I think of a number and divide it by 6. The answer is 72.
What is my number?



(d)

I think of a number and subtract 12 from it. The answer is 79.
What is my number?



(e)

I think of a number and add 36.
The answer is 78.
What is my number?



(f)

I think of a number, add 17 then subtract 6. The answer is 34.
What is my number?



2. Check each of your answers by putting the missing number into the question.



The **inverse** of an operation is the operation that has the opposite effect (for example, addition is the inverse of subtraction, and vice versa; multiplication is the inverse of division, and vice versa). You can use inverses to help you solve problems by working backwards from the answer: for example, if $a \times 9 = 27$, then $27 \div 9 = a$.